

Beam Resource Matching for Pilot Assignment

Combining Beam-Domain Conflict Modeling with Many-to-Many Matching

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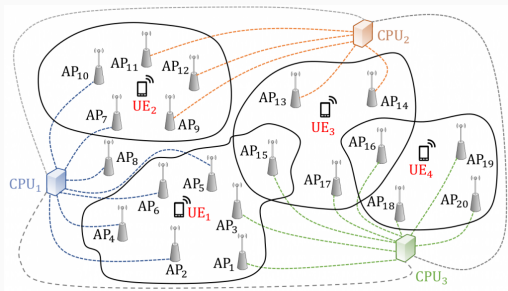
Mobile Communication Systems Term Project

System characteristic

- Many distributed APs jointly serve a smaller number of UEs ($L \gg K$), coordinated by a central processing unit (CPU) over fronthaul.
- No cell boundaries; coherent cooperative transmission yields handover-free coverage.

Why cell-free mMIMO

- Distributed APs shorten access distances and improve macro-diversity.
- Cell-free cooperation reduces cell-edge effects and SINR disparity.
- Dense AP-UE connectivity also creates many overlapping spatial links, making pilot reuse a central bottleneck.



Channel estimation via uplink pilots

A TDD coherence block of length τ_c allocates τ_p pilot symbols followed by $\tau_c - \tau_p$ data symbols. AP l receives

$$\mathbf{Y}_{p,l} = \sum_{k=1}^K \sqrt{\tau_p \rho_p} \mathbf{h}_{k,l} \phi_{p_k}^H + \mathbf{N}_{p,l},$$

where p_k is the pilot index assigned to UE k . Correlating with ϕ_{p_k} gives the observation used for MMSE channel estimation.

$$\text{SE}_k = \frac{\tau_c - \tau_p}{\tau_c} \log_2(1 + \text{SINR}_k).$$

Pilot contamination

- Typically $K > \tau_p$; UEs must reuse pilots.
- Same-pilot UEs corrupt one another's MMSE estimates.
- Contamination severity scales with the overlap of serving APs and beam directions among pilot-sharing UEs.

Design goal

Assign pilots so that same-pilot UEs share as little spatial structure as possible.

Gao (TVT 2024): AP-domain matching

- Group UEs by common serving APs.
- Many-to-many matching between UEs and APs.
- Orthogonal pilots within high-risk groups.

Limitation: AP-level matching does not resolve beam direction within a shared AP.

Mussbah (IEEE CL 2024): Beam-domain graph

- Beam-domain channel decomposition (DFT codebook).
- Active / moderate beam reports from initial access.
- Pilot assignment by graph coloring on beam overlap.

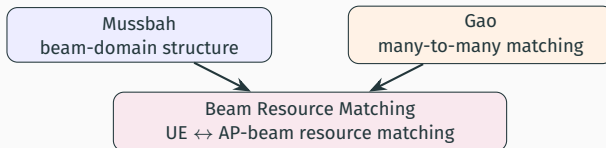
Limitation: dense beam-overlap graphs inflate the chromatic number and pilot overhead.

This work

Beam Resource Matching combines Gao's matching mechanism with Mussbah's beam-domain structure: many-to-many matching at the AP-beam resource level $r = (l, n)$.

Design principle

Beam Resource Matching lifts Gao's many-to-many matching from the AP domain to the AP-beam resource domain and inherits Mussbah's active/moderate beam structure.



- **Gao:** many-to-many matching between UEs and APs on large-scale fading.
- **Mussbah:** beam-domain conflict graph from active/moderate beam overlaps.
- **Beam Resource Matching:** matches UEs to resources $r = (l, n)$; the matching outcome determines the active beam set and feeds pilot assignment.

Resource definition

$$r = (l, n), \quad |\mathcal{R}| = LN.$$

- l : AP index.
- n : beam index at AP l .
- Each pair (l, n) constitutes a single matching resource.

UE-resource preference

$$P_{k,r} = P_{k,l,n}.$$

Resolution gain over Gao

- Gao ranks candidates by AP-domain strength $\beta_{k,l}$.
- Beam Resource Matching ranks by beam-domain power $P_{k,l,n}$.
- Beam-level resolution separates “same AP” from “same beam.”

Modeling distinction

Matching operates at the AP-beam resource level rather than the AP level.

Beam-domain channel

$$\mathbf{h}_{k,l}^{(B)} = \mathbf{U}^H \mathbf{h}_{k,l}.$$

UE k reports beam set $\mathcal{B}_k$. The strongest beams (cumulative power threshold) form the active set:

$$\mathcal{B}_k^{(a)} \subset \mathcal{B}_k.$$

Remaining reported beams form the moderate set:

$$\mathcal{B}_k^{(i)} = \mathcal{B}_k \setminus \mathcal{B}_k^{(a)}.$$

Beam-domain conflicts capture finer interference than AP-domain. Trade-off: dense graphs raise the chromatic number and inflate pilot overhead.

Conflict graph

$$\mathbf{B} = \mathbf{B}^{(a)T} \mathbf{B}^{(a)} + \mathbf{B}^{(a)T} \mathbf{B}^{(i)} + \mathbf{B}^{(i)T} \mathbf{B}^{(a)}.$$

$$A_{i,j} = 1 \quad \text{if } B_{i,j} > 0.$$

Pilots are assigned by graph coloring:

$$\tau_p^{\text{actual}} = \chi(G).$$

Design objective

Beam Weighted Threshold retains Mussbah's beam-domain coloring framework while suppressing weak edges through a weighted interference score.

$$B_{u,v}^w = w_{aa} |\mathcal{B}_u^{(a)} \cap \mathcal{B}_v^{(a)}| + w_{ai} |\mathcal{B}_u^{(a)} \cap \mathcal{B}_v^{(i)}| + w_{ia} |\mathcal{B}_u^{(i)} \cap \mathcal{B}_v^{(a)}|.$$

$$A_{u,v} = \begin{cases} 1, & B_{u,v}^w > \theta, u \neq v, \\ 0, & \text{otherwise.} \end{cases}$$

- w_{aa} emphasizes active-active overlap, the typically dominant interference.
- Threshold θ controls graph density.
- Pilots are assigned by graph coloring on G_θ :

$$\tau_p^{\text{actual}} = \chi(G_\theta).$$

Gao performs many-to-many matching between UEs and APs based on AP-domain large-scale fading $\beta_{m,k}$. The indicator

$$\mu(m, k) = 1$$

denotes that UE m is matched to AP k .

Quota constraints

$$|\mu(\text{UE}_m)| \leq Q_{\text{UE}}, \quad |\mu(\text{AP}_k)| \leq Q_{\text{AP}}.$$

Key setting:

$$Q_{\text{AP}} = \tau_{\text{p}}.$$

Limitation: AP-level matching cannot distinguish two UEs that share an AP through different beams.

AP group

$$\mathcal{G}_k = \{m : \mu(m, k) = 1\}.$$

Common serving AP ratio

$$M_{m_1, m_2}^{\text{AP}} = \frac{|\mu(m_1, :) \cap \mu(m_2, :)|}{|\mu(m_1, :)|}.$$

Group risk

$$S_k = \sum_{m_1 \neq m_2, m_1, m_2 \in \mathcal{G}_k} M_{m_1, m_2}^{\text{AP}}.$$

Sparse AP-domain conflict graph

$$\mathcal{T}_k(N_{\text{top}}) = \text{Top}_{N_{\text{top}}} \{\beta_{k,1}, \dots, \beta_{k,L}\}.$$

A conflict edge is placed when two UEs share at least one Top- N AP:

$$e_{i,j}^{\text{TopN}} = \mathbf{1}\{\mathcal{T}_i(N_{\text{top}}) \cap \mathcal{T}_j(N_{\text{top}}) \neq \emptyset\}.$$

Pilot assignment flow

- Identify the Top- N strongest APs per UE.
- Connect UEs whose Top- N sets intersect.
- Apply graph coloring on the resulting sparse graph.
- Connected UEs receive distinct pilots.

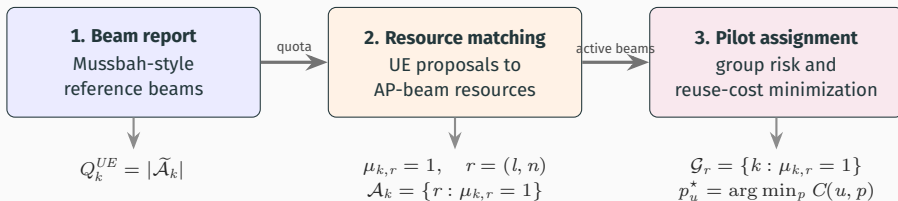
Algorithmic role

AP-Top- N provides a lightweight AP-domain sparsification of Gao's AP-overlap criterion.

Method	Basis	Assignment principle
Gao	AP-domain matching	Match UEs to APs and compute AP group risk
Beam Weighted Threshold	beam-domain	Create edges only when weighted beam overlap exceeds a threshold
Beam Resource Matching	AP-beam resource	Match UEs to AP-beam resources
AP-Top- N	AP-domain sparsification	Use only strongest AP Top- N overlap as conflict

Methodological position

AP-Top- N reuses Gao's large-scale AP information; Beam Resource Matching combines Mussbah's beam-domain structure with Gao's matching mechanism.



Operational mechanism

Beam reports set the UE quotas; matching determines the final active beam set; weighted overlap drives pilot assignment.

Reference quota

$$Q_k^{UE} = |\tilde{\mathcal{A}}_k|.$$

$\tilde{\mathcal{A}}_k$ denotes the reference active beam set from the Mussbah power-threshold rule.

Matching variable

$$\mu_{k,r} \in \{0, 1\}.$$

$$\mu_{k,r} = 1 \iff \text{UE } k \text{ is matched to resource } r.$$

Matching follows a deferred-acceptance protocol: UEs propose to their strongest resources, and each resource accepts up to Q_{res} UEs ranked by beam power.

Constraints

$$\sum_{r \in \mathcal{R}} \mu_{k,r} \leq Q_k^{UE}, \quad \sum_{k=1}^K \mu_{k,r} \leq Q_{\text{res}}.$$

$$\mu_{k,r} = 0 \quad \text{if } r \notin \mathcal{B}_k.$$

Typical choice:

$$Q_{\text{res}} = \tau_p^{\text{design}}.$$

Mussbah selects active beams via a per-UE power threshold.

Beam Resource Matching defines active beams as the matched resources:

$$\mathcal{A}_k = \{r : \mu_{k,r} = 1\}.$$

Unmatched reported beams form the moderate set:

$$\mathcal{I}_k = \mathcal{B}_k \setminus \mathcal{A}_k.$$

Algorithmic consequence

Active beams reflect global UE-resource competition, not only per-UE power ranking.

- Each beam must survive UE-resource competition.
- Resource quota Q_{res} caps UE concentration per AP-beam resource.
- RF-chain activity and pilot grouping are determined jointly by the matching outcome.

Each AP-beam resource induces a UE group from the matching outcome:

$$\mathcal{G}_r = \{k : \mu_{k,r} = 1\}.$$

Gao

AP group: $\mathcal{G}_k = \{m : \mu(m, k) = 1\}$.

Risk is computed inside each AP group.

Key replacement

Group structure shifts from AP groups (Gao) to AP-beam resource groups (Beam Resource Matching).

Beam Resource Matching

AP-beam resource group: $\mathcal{G}_r = \{k : \mu_{k,r} = 1\}$.

Risk is computed inside each resource group.

Pairwise interference between UE u and UE v is measured by a weighted overlap score:

$$W_{u,v} = w_{aa}|\mathcal{A}_u \cap \mathcal{A}_v| + w_{ai}|\mathcal{A}_u \cap \mathcal{I}_v| + w_{ia}|\mathcal{I}_u \cap \mathcal{A}_v|.$$

- $w_{aa}|\mathcal{A}_u \cap \mathcal{A}_v|$: shared active resources; dominant interference term.
- $w_{ai}|\mathcal{A}_u \cap \mathcal{I}_v|$: active for u , moderate for v ; secondary term.
- $w_{ia}|\mathcal{I}_u \cap \mathcal{A}_v|$: moderate for u , active for v ; secondary term.

Weight propagation

Weights propagate into the overlap ratio, resource group risk, and pilot reuse cost—not merely a diagnostic label.

Beam Resource Matching replaces Gao's common-serving-AP ratio with a beam-domain weighted counterpart:

$$M_{u,v}^B = \frac{W_{u,v}}{|\mathcal{A}_u|} = \frac{w_{aa}|\mathcal{A}_u \cap \mathcal{A}_v| + w_{ai}|\mathcal{A}_u \cap \mathcal{I}_v| + w_{ia}|\mathcal{I}_u \cap \mathcal{A}_v|}{|\mathcal{A}_u|}.$$

Unweighted baseline (Mussbah-style)

$$|\mathcal{A}_u \cap \mathcal{A}_v| + |\mathcal{A}_u \cap \mathcal{I}_v| + |\mathcal{I}_u \cap \mathcal{A}_v|.$$

All overlap types weighted equally.

Directed ratio

$$M_{u,v}^B \neq M_{v,u}^B$$

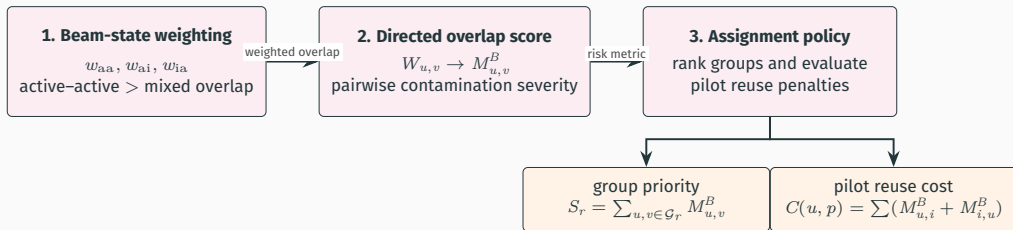
in general; denominators $|\mathcal{A}_u|$ and $|\mathcal{A}_v|$ typically differ.

Resource group risk aggregates the weighted beam-overlap ratio over UE pairs:

$$S_r = \sum_{u \neq v, u, v \in \mathcal{G}_r} M_{u,v}^B.$$

- Groups are processed in descending order of S_r .
- Within a group, unused pilots are assigned first.
- If no unused pilot remains, the pilot minimizing the weighted reuse cost is selected:

$$C(u, p) = \sum_{i: p_i = p} (M_{u,i}^B + M_{i,u}^B), \quad p_u^* = \arg \min_p C(u, p).$$



Assignment implication

The same weighted beam-overlap metric is used consistently for conflict scoring, group ordering, and pilot reuse selection.

The pilot count matches the largest resource group:

$$\tau_p^{\text{BRM}} = \max_r |\mathcal{G}_r|.$$

The matching stage enforces

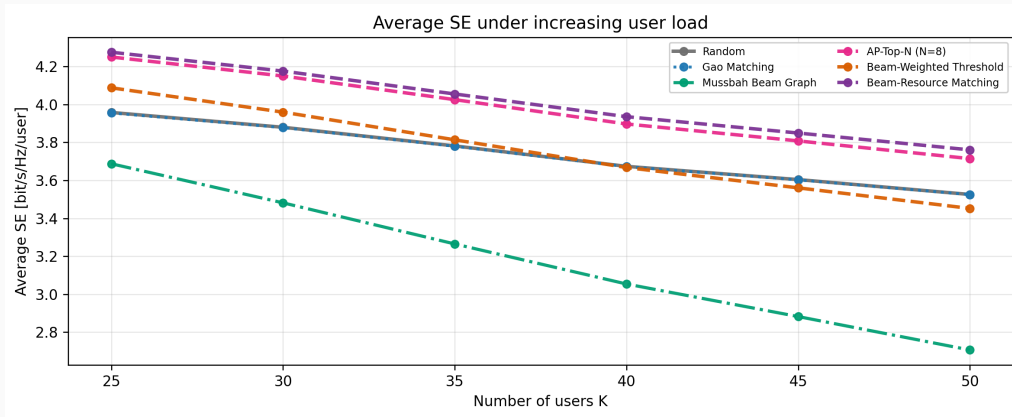
$$|\mathcal{G}_r| \leq Q_{\text{res}} = \tau_p^{\text{design}},$$

so the adaptive count is upper-bounded by the design budget:

$$\tau_p^{\text{BRM}} \leq \tau_p^{\text{design}}.$$

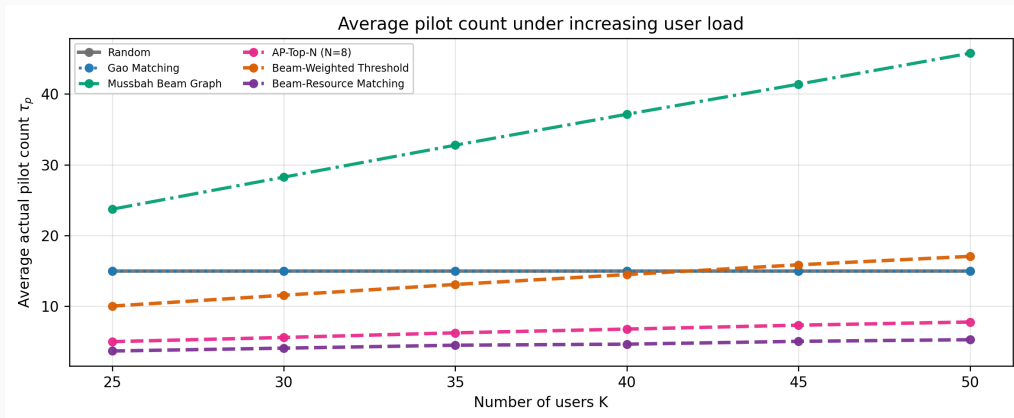
Pilot-count control

Mussbah's chromatic number grows unbounded with graph density. Beam Resource Matching caps the pilot count at the resource quota $Q_{\text{res}} = \tau_p^{\text{design}}$.



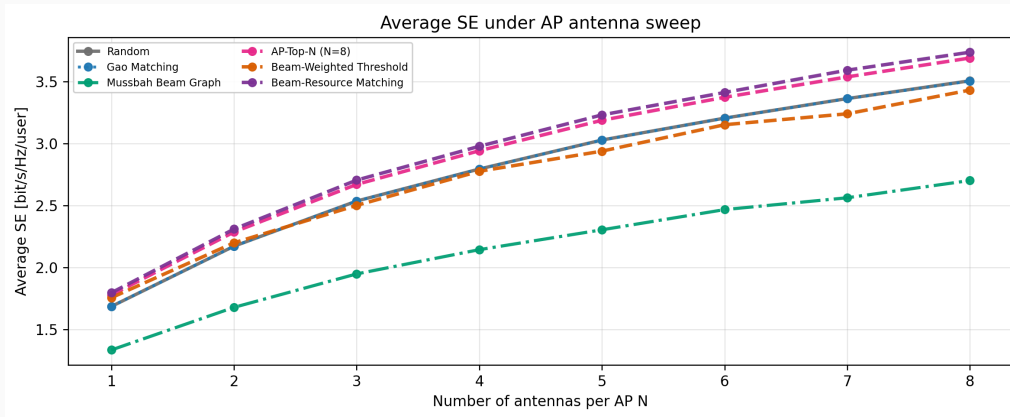
Configuration: $L = 200$, $N = 8$, $K = 25-50$, $\tau_c = 150$, $\tau_p^{\text{design}} = 15$, 200 setups.

Observation at $K = 50$: Beam Resource Matching +6.66% mean SE, AP-Top- N +5.35%; Gao remains close to Random, while Mussbah is penalized by inflated τ_p .



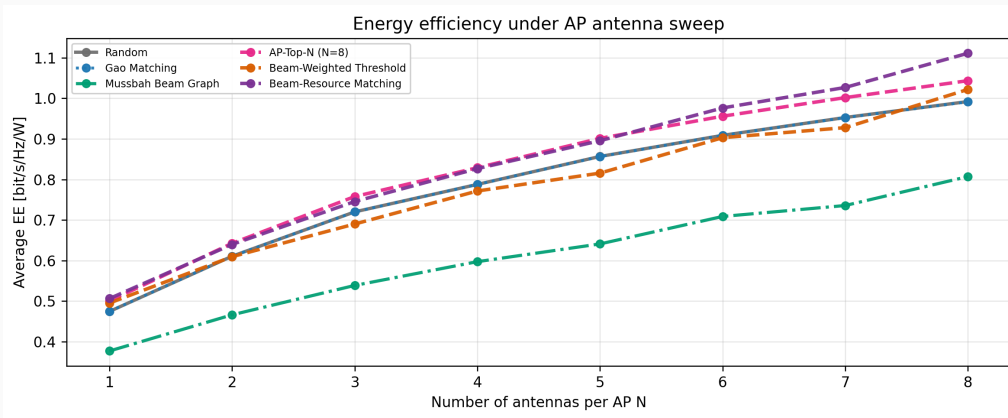
Pilot-overhead profile: Random/Gao use the fixed design budget, Mussbah's chromatic number grows rapidly, and Beam Resource Matching stays below the design budget.

At $K = 50$: Random/Gao $\tau_p = 15$, AP-Top-N $\tau_p \approx 7.79$, Beam Resource Matching $\tau_p \approx 5.29$, Mussbah $\tau_p \approx 45.76$.



Configuration: $L = 200$, $K = 50$, $N = 1, \dots, 8$, $\tau_c = 150$, $\tau_p^{\text{design}} = 15$, 200 setups.

Observed trend: Beam Resource Matching attains the highest mean SE across the tested antenna range; at $N = 8$, the gain is +6.57%.



Configuration: $L = 200$, $K = 50$, $N = 1, \dots, 8$, $\tau_c = 150$, $\tau_p^{\text{design}} = 15$, 200 setups.

Observed trend: Beam Resource Matching leads the EE proxy across the tested antenna range; at $N = 8$, the gain is +12.01%.

1. Beam-domain is more precise

same AP $\neq$ same beam.

Two UEs can share an AP while occupying different beams.

2. Active beams are matched

- Active beams emerge from global UE-resource competition, not per-UE power alone.
- Resource quota Q_{res} bounds UE concentration per resource.

3. Conflict severity is weighted

$$M_{u,v}^B = \frac{w_{aa}|\mathcal{A}_u \cap \mathcal{A}_v| + w_{ai}|\mathcal{A}_u \cap \mathcal{I}_v| + w_{ia}|\mathcal{I}_u \cap \mathcal{A}_v|}{|\mathcal{A}_u|}.$$

Strong beam-domain collisions are prioritized during pilot assignment.

Spectral efficiency contains a pilot-overhead prelog factor:

$$SE_k = \frac{\tau_c - \tau_p}{\tau_c} \log_2(1 + \text{SINR}_k).$$

- Reducing τ_p raises the prelog factor; SE increases provided SINR is preserved.
- A smaller active beam set reduces RF-chain activity.
- Combined effect lowers total active power and improves the EE proxy.

Design implication

Beam Resource Matching jointly reduces pilot overhead and RF-chain activity while preserving the dominant beam-domain interference structure.

Beam Resource Matching

Beam Resource Matching extends Gao's AP-domain many-to-many matching to Mussbah's beam-domain AP-beam resources.

- UEs are matched to resources $r = (l, n)$, each representing one AP-beam pair.
- The matching outcome defines the final active beam set:

$$\mathcal{A}_k = \{r : \mu_{k,r} = 1\}.$$

- The weighted ratio $M_{u,v}^B$ drives resource group risk S_r and pilot reuse cost $C(u, p)$.
- The actual pilot count is bounded by the resource quota:

$$\tau_p = \max_r |\mathcal{G}_r| \leq \tau_p^{\text{design}}.$$

Core principle: prioritize dominant beam-domain collisions while bounding pilot overhead through resource quotas.